

Competing Probabilistic Cognitive Models

Benchmark Analysis & Mathematical Formulations for 2AFC Reversal Tasks

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Abstract

This report provides the formal mathematical definitions, R implementation code, parameter estimation results, and model comparison metrics for discrete-time cognitive baselines benchmarked against the **ExactRModel** continuous-time cerebellar reservoir. Evaluated on 16,029 empirical human trials across 128 subjects, we benchmark: 1) a Counterfactual Expected Value (EV) Rescorla-Wagner model, and 2) a Win-Stay / Lose-Shift (WSLS) Markov heuristic model. Additionally, we propose two advanced Bayesian architectures: a Hidden Markov Model (HMM) and the Hierarchical Gaussian Filter (HGF).

1 Model Formulations & Mathematical Update Rules

1.1 Model A: Counterfactual Expected Value Rescorla-Wagner (EV-RW)

Because reward magnitudes (M_A, M_B) are explicitly visible to participants, the agent tracks internal probability estimates $\hat{p}_t(a) \in [0, 1]$ rather than raw Q-values.

When option $a_t \in \{1, 2\}$ is chosen and outcome $F_t \in \{0, 1\}$ is received:

1. Direct Prediction Error Update (Chosen Option a_t):

$$\delta_t = F_t - \hat{p}_t(a_t) \quad (1)$$

$$\hat{p}_{t+1}(a_t) = \hat{p}_t(a_t) + \alpha \cdot \delta_t \quad (2)$$

where $\alpha \in [0, 1]$ is the direct learning rate.

2. Counterfactual Prediction Error Update (Unchosen Option $\neg a_t$):

$$\hat{p}_{t+1}(\neg a_t) = \hat{p}_t(\neg a_t) + \alpha_c \cdot [(1 - F_t) - \hat{p}_t(\neg a_t)] \quad (3)$$

where $\alpha_c \in [0, 1]$ is the counterfactual learning rate.

3. Expected Value Computation:

$$EV_A(t) = \hat{p}_t(1) \times M_A(t), \quad EV_B(t) = \hat{p}_t(2) \times M_B(t) \quad (4)$$

4. Softmax Action Selection:

$$P(\text{Choice}_t = 1) = \frac{1}{1 + \exp(-\beta_{\text{softmax}} \cdot [EV_A(t) - EV_B(t)])} \quad (5)$$

1.2 Model B: Win-Stay / Lose-Shift (WSLS) Markov Heuristic

The WSLS baseline relies on a 2-parameter heuristic transition rule:

$$P(\text{Choice}_t = \text{Choice}_{t-1} \mid F_{t-1} = 1) = P_{\text{win_stay}} \quad (6)$$

$$P(\text{Choice}_t \neq \text{Choice}_{t-1} \mid F_{t-1} = 0) = P_{\text{lose_shift}} \quad (7)$$

2 Empirical Fitting Results & Model Benchmarks

Both models were fitted via Maximum Likelihood Estimation (`optim`) across all 128 human subjects.

Table 1: Empirical Model Fitting & Comparison Metrics (128 Human Participants, 16,029 Trials).

Model Architecture	Free Params	Mean NLL	Total AIC	Total BIC	Key Fitted Parameters
Counterfactual EV-RW	3	58.66	15,783.8	16,870.3	$\alpha = 0.4769, \alpha_c = 0.1775, \beta = 1$
Win-Stay Lose-Shift	2	53.25	14,144.9	14,867.2	$P_{\text{ws}} = 0.8736, P_{\text{ls}} = 0.5400$

3 Advanced Bayesian Expansion Proposals

To capture reversal dynamics better than standard RL, we propose two advanced extensions:

3.1 Expansion 1: Hidden Markov Model (HMM) Latent State Tracker

Maintains a belief vector $\mathbf{b}_t = [P(S_t = 1), P(S_t = 2)]^T$ over latent reversal state $S_t \in \{1, 2\}$, updating beliefs via transition matrix $\mathbf{T} = \begin{bmatrix} 1 - \gamma & \gamma \\ \gamma & 1 - \gamma \end{bmatrix}$ where $\gamma \ll 0.1$ is the expected reversal probability.

3.2 Expansion 2: Hierarchical Gaussian Filter (HGF)

A 3-level Bayesian filter tracking environmental volatility ϑ_t . Precision-weighted prediction errors dynamically adjust learning rates $\alpha(t)$, accelerating belief updates immediately following a reversal.